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|---------------------------|--------------------------------------------|------------------------|-------------|
| <b>Course Title</b>       | Power Electronics                          | <b>Course Code</b>     | EE3032      |
| <b>Department</b>         | Department of Electrical Engineering (DEE) | <b>Campus</b>          | Lahore      |
| <b>Knowledge Profile</b>  | Engineering Specialization (WK4)           | <b>Credit Hrs.</b>     | 3+1         |
| <b>Knowledge Area</b>     | Electronics (KA04)                         | <b>Grading Scheme</b>  | Relative    |
| <b>HEC Knowledge Area</b> | Breadth Core                               | <b>Applicable From</b> | Spring 2023 |
| <b>SDG</b>                | 7   Affordable and Clean Energy            | <b>PBL</b>             | 1           |
| <b>Pre-requisite(s)</b>   | EE/EL1004 Electronic Devices & Circuits    | <b>CEP</b>             | 1           |

|                         |                                                                                                                                                |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Course Objective</b> | This course guides students through the techniques required in the design, analysis, and fabrication of power electronic circuits and systems. |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|

| No. | Assigned Program Learning Outcome (PLO)                                                                                                                                                                                                                  |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2   | An ability to identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.                                    |
| 3   | An ability to design solutions for complex engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations. |

I = Introduction, R = Reinforcement, E = Evaluation, A = Assignment, Q = Quiz, M = Midterm, F=Final, L = Lab, P = Project, W = Written Report.

| No. | Course Learning Outcome (CLO) Statements      | Assessment Tools | Taxonomy Levels | PLO |
|-----|-----------------------------------------------|------------------|-----------------|-----|
| 1   | Analyze op-amp application in power circuits. | Q1, M1           | C4              | 2   |
| 2   | Analyze audio power amplifier circuits.       | A1, M1           | C4              | 2   |
| 3   | Design power switches.                        | Q2, M2, F        | C5              | 3   |
| 4   | Design power converter circuits.              | Q3, F            | C5              | 3   |
| 5   | Analyze thyristor-based circuits              | A2, F            | C4              | 2   |

|                        |                  |                                              |
|------------------------|------------------|----------------------------------------------|
| <b>Text Books</b>      | <b>Title</b>     | Power Electronics, Principles & Applications |
|                        | <b>Author</b>    | J. Michael Jacob                             |
|                        | <b>Publisher</b> | © 2002 Thomson Delmar                        |
| <b>Reference Books</b> | <b>Title</b>     | Power Electronics                            |
|                        | <b>Author</b>    | Ned Mohan                                    |
|                        | <b>Publisher</b> |                                              |

| Week  | Course Contents/Topics                                                                          | Chapter* | CLO* |
|-------|-------------------------------------------------------------------------------------------------|----------|------|
| 01    | Advanced Operational Amplifier Principles, Op Amp Characteristics, Universal Analysis Technique | 1        | 1    |
| 02    | Composite Amplifier, Suspended Supply Amplifier                                                 | 1        | 1    |
| 03    | Power Parameter Calculations, Common Waveforms, Average value, Root Mean Square Value, Power    | 3        | 1    |
| 04    | Linear Power Amplifier                                                                          | 4        | 2    |
| 05    | Low Power Audio Amplifier, High Power Audio Amplifier                                           | 4        | 2    |
| 06    | The Enhancement Mode MOSFET, Class A Common Drain Amplifier,                                    | 5        | 2    |
| 07    | Class B Push-Pull Amplifier, Class B Amp with Op Amp Driver                                     | 5        | 2    |
| 08    | Parallel MOSFET in a Linear Amp, Amplifier Protection, Driving a Reactive Load                  | 5        | 2    |
| 09    | Power Switches, Switching Characteristics (Diodes, Transistors), Parallel MOSFET Switches       | 6        | 3    |
| 10    | Low Side Switches, High Side Switches, MOSFET Switch Drivers, H Bridge                          | 6        | 3    |
| 11    | Switching Power Supplies, Buck Regulators                                                       | 7        | 4    |
| 12-13 | Boost Regulators, Line Voltage Flyback Converter, AC-to-DC Converters                           | 7        | 4    |
| 14    | Thyristors, Device Characteristics                                                              | 8        | 5    |
| 15    | Snubbing, Triggers, Proportional Power Circuits                                                 | 8        | 5    |
| 16    | Three Phase Converters                                                                          | 9        | 4    |

\*Reference book chapters are given in brackets

| Assessment Tools      | Weightage |
|-----------------------|-----------|
| Quizzes + Assignments | 20.0%     |
| Midterms (I+II)       | 30.0%     |
| Final Exam            | 50.0%     |